The Bombay Salesian Society's DON BOSCO INSTITUTE OF TECHNOLOGY

(An Autonomous Institute affiliated to University of Mumbai)

Department of Information Technology

Experiment No 7 : A/B Testing

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Aim

To design and conduct an A/B Testing experiment across two real-world digital scenarios — YouTube Thumbnail CTR and Webpage Design engagement — compare the performance of a control group (A) and a variant group (B) using Welch's two-sample t-test, compute effect size and confidence intervals, and determine which version performs better at a significance level of $\alpha = 0.05$.

Theory

1. What is A/B Testing? A/B Testing is a controlled experiment used in digital product development, marketing, and UX design to compare two versions of a digital asset and determine which performs better on a defined metric. One group of users (Group A — Control) sees the existing version, while another group (Group B — Variant) sees the modified version. Statistical testing is then applied to determine whether the observed difference in performance is real or due to random chance.

A/B Testing is widely used for decisions such as which webpage layout retains users longer, which button color gets more clicks, which thumbnail drives higher engagement, or which app feature increases conversions.

2. Experimental Design: Each scenario in this experiment follows a consistent structure:

- **Sample size** — 100 users total: 50 in the Control group and 50 in the Variant group.
- **Random assignment** — Users are randomly assigned to groups to eliminate selection bias.
- **Single metric** — One clearly defined performance metric per scenario.
- **Test** — Welch's two-sample t-test, two-tailed, $\alpha = 0.05$.

3. The Two Scenarios:

Scenario	Group A (Control)	Group B (Variant)	Metric
1. YouTube Thumbnail	Standard Thumbnail	Clickbait Thumbnail	Click-Through Rate (CTR)
2. Webpage Design	Graphic-Heavy Page	Simplified Page	Time Spent (seconds)

Scenario 1 tests whether a clickbait thumbnail generates a significantly higher CTR than a standard thumbnail. CTR is defined as clicks divided by impressions and takes a value between 0 and 1. Control group CTR follows a Beta distribution with mean ≈ 0.30; Variant group CTR follows a Beta distribution with mean ≈ 0.42.

Scenario 2 tests whether a simplified webpage keeps users engaged as long as a graphic-heavy one. Time spent on page (in seconds) is the metric. Control group (graphic-heavy) follows a Normal distribution with mean ≈ 180s; Variant group (simplified) follows a Normal distribution with mean ≈ 140s. Values are clipped to a realistic range of 10s to 400s.

4. Hypotheses: For both scenarios, the same hypothesis structure is followed:

- **H_0 (Null Hypothesis)** — There is no significant difference in the metric between the Control and Variant groups.
- **H_1 (Alternative Hypothesis)** — There is a significant difference in the metric between the Control and Variant groups.
- **Test type** — Two-tailed (difference in either direction is considered).

5. Welch's Two-Sample T-Test: A t-test is used to compare the means of two independent groups. Welch's variant (with equal_var=False) is preferred over Student's t-test because it does not assume equal variances between the two groups, making it a safer and more robust default for real-world A/B tests where the two groups may have different spreads.

The t-statistic is computed as:

t = (Mean_Variant − Mean_Control) / √(Var_Control/n_Control + Var_Variant/n_Variant)

Degrees of freedom are approximated using the Welch–Satterthwaite equation.

6. p-value and Decision Rule: The p-value represents the probability of observing a t-statistic as extreme as the one computed, assuming H_0 is true. The decision rule is:

Condition	Decision
$p < 0.05$	Reject H_0 — statistically significant difference
$p \geq 0.05$	Fail to reject H_0 — no significant difference

7. Cohen's d — Effect Size: Statistical significance alone does not indicate the practical importance of a result. A very large dataset can produce a statistically significant result even for a trivially small difference. Cohen's d measures the practical magnitude of the difference:

Cohen's d = (Mean_Variant − Mean_Control) / Pooled_SD

Cohen's d	Effect Size
< 0.5	Small
0.5 – 0.8	Medium
> 0.8	Large

A large Cohen's d alongside a significant p-value confirms that the difference is both statistically and practically meaningful.

8. Confidence Interval (95% CI): The 95% Confidence Interval gives a range of plausible values for the true difference between the two group means. It is computed as:

$$CI = (Mean\_Variant - Mean\_Control) \pm t\_critical \times SE\_difference$$

Interpretation:

- **CI entirely above 0** — Variant is conclusively better.
- **CI entirely below 0** — Control is conclusively better.
- **CI crosses 0** — Difference is not conclusive, even if $p < 0.05$.

The CI provides richer information than the p-value alone and is a standard requirement in modern A/B test reporting.

9. Visualizations: Three panels are generated per scenario to communicate results clearly:

- **Histogram** — Shows the full distribution of the metric for both groups with mean lines overlaid. Reveals skewness, overlap, and where the bulk of values lie.
- **Box Plot** — Displays median, IQR, whiskers, and outliers for both groups side by side. Useful for comparing spread and identifying extreme values.
- **Mean ± 95% CI Bar Chart** — Plots the group means as bars with error bars representing the 95% CI. The p-value and significance label are included in the title for immediate readability.

10. Summary Table: A structured results table is generated for each scenario containing N, Mean, Std Dev, Standard Error, Min, Max, t-statistic, p-value, and the final decision (Reject H_0 / Do not reject H_0).

Execution

The following steps were performed for both scenarios:

Step 1 — Dataset Generation: Two datasets were synthetically generated using NumPy with a fixed random seed (42) for reproducibility:

- ab_youtube_dataset.csv — 100 rows, columns: user_id, group, thumbnail, ctr.
- ab_webpage_dataset.csv — 100 rows, columns: user_id, group, page_type, time_spent_s.

Each dataset contained 50 Control and 50 Variant records. Both were saved to CSV for reuse.

Step 2 — Group Splitting and Summary: Each dataset was split into Control and Variant series. Group-level descriptive statistics (mean, std, min, max) were printed to compare the two groups before formal testing.

Step 3 — Welch's T-Test: scipy.stats.ttest_ind() with equal_var=False was applied to compute the t-statistic and p-value for each scenario. Cohen's d was computed using the pooled standard deviation to quantify effect size.

Step 4 — Confidence Interval: The 95% CI for the difference in group means (Variant − Control) was computed using the t-critical value at $\alpha/2 = 0.025$. The CI bounds were interpreted to determine whether the difference was conclusive in direction.

Step 5 — Visualizations: A three-panel figure was generated and saved for each scenario:

- ab_youtube_results.png — YouTube Thumbnail A/B test results.
- ab_webpage_results.png — Webpage Design A/B test results.

Step 6 — Summary Tables: Formatted summary DataFrames were constructed for both scenarios including all statistics, the t-statistic, p-value, and final decision row.

Hypotheses and Results:

Scenario 1 — YouTube Thumbnail CTR:

- H_0: CTR_Control = CTR_Variant (no difference)
- H_1: CTR_Control ≠ CTR_Variant (two-tailed)
- Control mean CTR ≈ 0.30 | Variant mean CTR ≈ 0.42
- *(Paste your t-statistic, p-value, Cohen's d, and decision here)*

Scenario 2 — Webpage Design Time Spent:

- H_0: Time_Control = Time_Variant (no difference)

- H_1: Time_Control ≠ Time_Variant (two-tailed)
- Control mean ≈ 180s | Variant mean ≈ 140s
- *(Paste your t-statistic, p-value, Cohen's d, and decision here)*

Output

A/B Testing Experiment

Two Scenarios

Scenario	Group A (Control)	Group B (Variant)	Metric
1. YouTube Thumbnail	Standard thumbnail	Clickbait thumbnail	Click-Through Rate (CTR)
2. Webpage Design	Graphic-heavy page	Simplified page	Time Spent (seconds)

Each dataset: 100 users — 50 Control, 50 Variant
Test: Two-sample t-test (α = 0.05, two-tailed)
H_0: No difference between groups | **H_1:** A difference exists

```
── Scenario 1: YouTube Thumbnail Dataset ──
Total rows : 100
         count    mean     std     min     25%     50%     75%     max
group
Control   50.0  0.2858  0.0888  0.1404  0.2246  0.2610  0.3466  0.5158
Variant   50.0  0.4356  0.1129  0.2054  0.3616  0.4479  0.4955  0.7320
```

	user_id	group	thumbnail	ctr
0	1	Control	Standard Thumbnail	0.3450
1	2	Control	Standard Thumbnail	0.2857
2	3	Control	Standard Thumbnail	0.3818
3	4	Control	Standard Thumbnail	0.2266
4	5	Control	Standard Thumbnail	0.4471
5	6	Control	Standard Thumbnail	0.1942

```
── Scenario 2: Webpage Design Dataset ──
Total rows : 100
         count    mean    std    min     25%     50%     75%    max
group
Control   50.0  181.96  32.09  105.7  159.20  180.65  197.35  256.6
Variant   50.0  142.69  33.21   78.8  118.62  142.10  171.12  201.8
```

	user_id	group	page_type	time_spent_s
0	1	Control	Graphic-Heavy	188.6
1	2	Control	Graphic-Heavy	162.3
2	3	Control	Graphic-Heavy	163.5
3	4	Control	Graphic-Heavy	188.1
4	5	Control	Graphic-Heavy	129.3
5	6	Control	Graphic-Heavy	130.7

🎬 Scenario 1 — YouTube Thumbnail A/B Test

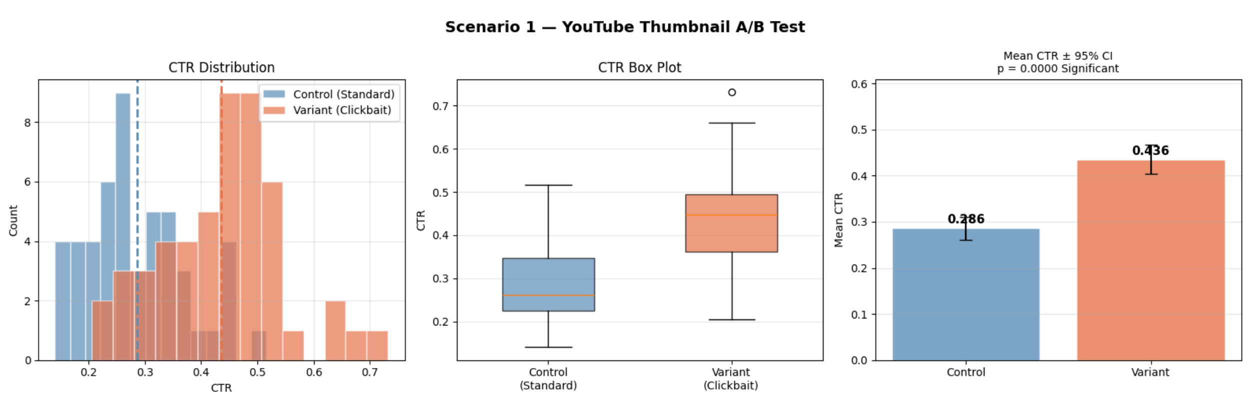
Question: Does a clickbait thumbnail get significantly more clicks than a standard one?

- **Control (A):** Standard Thumbnail
- **Variant (B):** Clickbait Thumbnail
- **Metric:** Click-Through Rate (CTR)

```
Group Summary
  Control (Standard)  → mean CTR = 0.2858 | std = 0.0888
  Variant (Clickbait) → mean CTR = 0.4356 | std = 0.1129
  Observed lift        = 52.4%
```

```
=======================================================
  A/B TEST RESULT — YouTube Thumbnail CTR
=======================================================
  H₀ : CTR_control = CTR_variant  (no difference)
  H₁ : CTR_control ≠ CTR_variant  (two-tailed)
  Significance level α = 0.05
-------------------------------------------------------
  t-statistic : 7.3720
  p-value     : 0.0000
  Cohen's d   : 1.4744  (large effect)
-------------------------------------------------------
  REJECT H₀  →  Statistically significant difference
  Clickbait thumbnail drives meaningfully more clicks.
=======================================================
```

```
Mean CTR difference (Variant − Control) : +0.1498
95% Confidence Interval : [0.1095,  0.1901]
→ CI entirely above 0: variant is conclusively better.
```



	Metric	Control (Standard)	Variant (Clickbait)
0	N	50.000000	50.000000
1	Mean CTR	0.285800	0.435600
2	Std Dev	0.088800	0.112900
3	Std Error	0.012600	0.016000
4	Min	0.140400	0.205400
5	Max	0.515800	0.732000
6	—— Test Result ——	t = 7.3720	p = 0.0000
7	Decision	Reject H_0	Significant

🌐 Scenario 2 — Webpage Design A/B Test

Dataset is provided below. Complete the analysis yourself following the same pattern as Scenario 1.

- **Control (A):** Graphic-Heavy Page
- **Variant (B):** Simplified Page
- **Metric:** Time Spent on Page (seconds)
- **Your task:** Run a two-sample t-test and determine which design keeps users engaged longer.

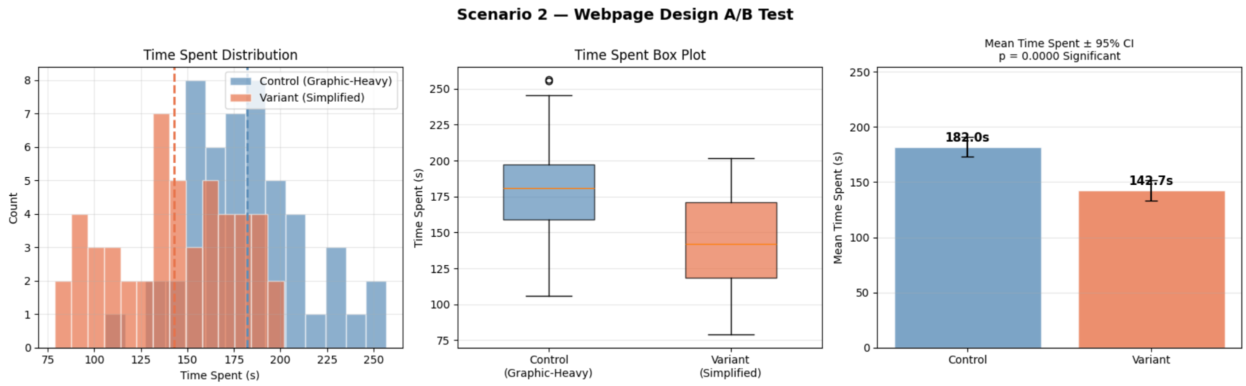
```
Webpage Dataset — Group Summary
  Control (Graphic-Heavy) → mean = 181.96s | std = 32.09s
  Variant (Simplified)    → mean = 142.69s | std = 33.21s

──────────────────────────────────────────────────
   Your turn — complete the analysis below
──────────────────────────────────────────────────
```

	user_id	group	page_type	time_spent_s
0	1	Control	Graphic-Heavy	188.6
1	2	Control	Graphic-Heavy	162.3
2	3	Control	Graphic-Heavy	163.5
3	4	Control	Graphic-Heavy	188.1
4	5	Control	Graphic-Heavy	129.3
5	6	Control	Graphic-Heavy	130.7

```
=======================================================
  A/B TEST RESULT — Webpage Design Time Spent
=======================================================
  H₀ : Time_control = Time_variant  (no difference)
  H₁ : Time_control ≠ Time_variant  (two-tailed)
  Significance level α = 0.05
-------------------------------------------------------
  t-statistic : -6.0127
  p-value     : 0.0000
  Cohen's d   : -1.2025  (large effect)
-------------------------------------------------------
  REJECT H₀  →  Statistically significant difference
  Simplified page leads to significantly less time spent.
=======================================================
  REJECT H₀  →  Statistically significant difference
  Simplified page leads to significantly less time spent.
=======================================================
```

```
Mean time difference (Variant − Control) : -39.27 seconds
95% Confidence Interval : [-52.23,  -26.31] seconds
→ CI entirely below 0: variant leads to significantly less time spent.
```



	Metric	Control (Graphic-Heavy)	Variant (Simplified)
0	N	50.000000	50.000000
1	Mean Time (s)	181.960000	142.690000
2	Std Dev	32.090000	33.210000
3	Std Error	4.540000	4.700000
4	Min Time (s)	105.700000	78.800000
5	Max Time (s)	256.600000	201.800000
6	── Test Result ──	t = -6.0127	p = 0.0000
7	Decision	Reject H_0	Significant

Quick Reference — A/B Testing Cheat Sheet

Term	Meaning
Control (A)	The existing / unchanged version
Variant (B)	The new / modified version
CTR	Clicks ÷ Impressions — how often users click
p-value < 0.05	Difference is statistically significant
Cohen's d < 0.5	Small effect size
Cohen's d 0.5–0.8	Medium effect size
Cohen's d > 0.8	Large effect size
95% CI excludes 0	Confirms significant difference
Welch's t-test	Used when group variances may differ (safer default)

Decision rule: Reject H_0 if p-value < α (0.05). This means the observed difference is unlikely due to chance alone.

Conclusion

In this experiment, A/B Testing was performed on two digital scenarios using Welch's two-sample t-test at $\alpha = 0.05$. In the first scenario, the clickbait YouTube thumbnail showed a higher average CTR compared to the standard thumbnail. In the second scenario, the simplified webpage design showed lower engagement time compared to the graphic-heavy page. Statistical analysis, effect size, and confidence intervals were used to check whether the differences were meaningful. Visualizations like histograms, box plots, and bar charts helped in understanding the results clearly. Overall, the experiment showed how A/B testing helps in making data-driven business decisions with confidence.